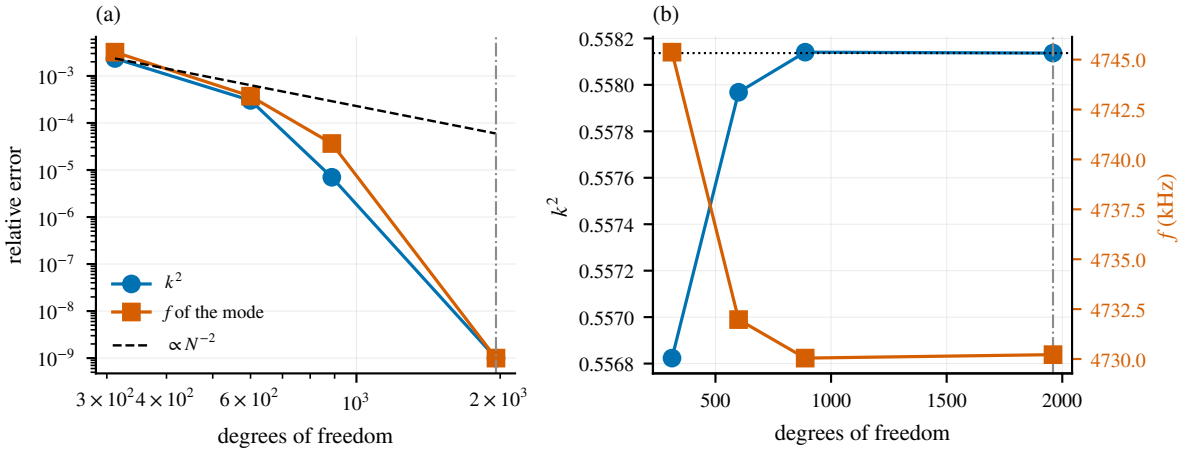
(a)
relative error
$10^{-3}$
$10^{-4}$
$10^{-5}$
$10^{-6}$
$10^{-7}$
$10^{-8}$
$10^{-9}$
$k^2$
$f$ of the mode
$\propto N^{-2}$
$3\times10^2$
$4\times10^2$
$6\times10^2$
$10^3$
$2\times10^3$
degrees of freedom
(b)
$k^2$
0.5582
0.5580
0.5578
0.5576
0.5574
0.5572
0.5570
0.5568
4745.0
4742.5
4740.0
4737.5
4735.0
4732.5
4730.0
$f$ (kHz)
500
1000
1500
2000
degrees of freedom